BSIDES SLC | WORKSHOP | 2 HOURS

Cybersecurity Careers Without the Hype.

A hands-on workshop for people who want to break into cybersecurity and actually get hired.

Rupinder Pal Singh

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CISSP | CISM | CISA | CRISC

April 10, 2026 | Student Event of the Year

Workshop Roadmap

2 hours. 8 blocks. 4 hands-on exercises. You leave with deliverables.

01	The Reality Check	Why cyber matters, the 2026 job market, and what companies actually need	0:00
02	The Role Landscape	Role Family Discovery Quiz + 30+ roles + day-in-the-life	0:15
03	AI + The SOC Analyst Future	What AI is automating, what it can't, and your learning path	0:35
04	GRC: The Underrated Path	Deep dive into governance, risk, compliance + framework comparison	0:50
05	Build in Public	What to create and share for each role family	1:05
06	BREAK	Stand up, stretch, find a new partner	1:15
07	Getting Hired	Resumes, interviews, ATS, LinkedIn + live resume workshop	1:20
08	Mindset + Next Steps	Imposter syndrome, 30-day sprint, accountability partners	1:45

Why Cybersecurity Matters

It's not just about hackers. It's about trust.

$4.88M

Average cost of a data breach (2024)

514K+

Open cybersecurity jobs in the U.S.

33%

Projected job growth through 2033

3.5M

Global cybersecurity workforce shortage

Every company is a technology company. Every technology company needs cybersecurity. The question is not whether the field is growing. The question is where YOU fit.

Sources: IBM Cost of a Data Breach 2024, ISC2 Workforce Study, BLS Projections

Why Companies Hire Us

The business problems we are paid to solve

Protect Revenue

A single breach costs $4.4M on average ($10.2M in the US). You are hired to prevent losses that dwarf your salary.

Regulatory Compliance

FedRAMP, SOC 2, PCI DSS, HIPAA, GDPR. Non-compliance means fines, lost contracts, and blocked revenue.

Customer Trust

One breach can destroy years of brand reputation. You protect the trust that keeps customers paying and staying.

Enable the Business

Security that blocks everything kills productivity. Companies need people who can say yes, safely and move fast.

Defend Against AI & Cyber Threats

97% of orgs with security incidents lacked proper controls. New attack surfaces demand new defenders.

Reduce Risk & Insurance

Cyber insurance premiums are rising. Strong security programs lower premiums and satisfy board-level risk appetite.

The 2026 Job Market: Real Numbers

The Good News

- Cybersecurity grows 7x faster than the overall job market
- 514,000+ open positions in the U.S. alone
- AI security roles grew 40% year-over-year
- GRC demand surging: NIS2, DORA, SEC rules, EU AI Act
- 48% of newcomers enter at age 39+ (career changers welcome)
- AI-savvy professionals earn 22% more than peers

The Real Talk

- Entry-level is fiercely competitive (hundreds of applicants per role)
- Tier 1 SOC analyst postings dropped 53% since 2022
- "3-5 years experience" appears on most "entry-level" posts
- AI is reshaping which skills matter (and which don't)
- Certifications & Degree alone won't get you hired anymore
- You need to SHOW your work, not just list credentials

This workshop gives you a plan for both realities.

Four Families. Every Cybersecurity Role.

Find which family matches your energy, then explore the roles inside it.

BUILDERS

Security Engineer
Cloud Security Engineer
Security Architect
AppSec Engineer
DevSecOps Engineer
IAM Engineer

"I want to build secure systems."

BREAKERS

Penetration Tester
Red Team Operator
Vulnerability Analyst
Bug Bounty Researcher
AI Red Teamer

"I want to find what's broken and where the gaps are."

DEFENDERS

SOC Analyst
Incident Responder
Detection Engineer
Threat Intel Analyst
Forensics Analyst
NOC Analyst

"I want to protect and respond."

GOVERNORS

GRC Analyst
IT Auditor
Risk Analyst
Compliance Manager
Privacy Analyst/DPO
AI Governance Specialist

"I want to ensure accountability."

Where Every Role Lives

9 defense layers. 40+ roles. One interactive map.

01	**Customers** Client devices, browsers, mobile apps	4 roles
02	**Internet** DNS, CDN, DDoS protection	3 roles
03	**Perimeter** Firewalls, WAF, IDS/IPS, VPN	5 roles
04	**DMZ** Proxy servers, mail gateways, web servers	4 roles
05	**Internal Network** Switches, routers, segmentation, VLAN	6 roles
06	**Cloud Infrastructure** AWS/Azure/GCP, IAM, security groups	5 roles
07	**Employees** Endpoints, Active Directory, awareness	5 roles
08	**GRC** Policy, compliance, risk, audit	7 roles
09	**Incident Response** SIEM, SOAR, forensics, recovery	4 roles

LIVE INTERACTIVE DEMO

Defense in Depth Map

Click any layer or security device to see which roles protect it, what they earn, and how to get there.

Interactive Defense in Depth Map

WHAT YOU WILL SEE

01 Left-to-right data flow from customers through the internet into the company boundary

02 Click any security device (Firewall, WAF, SIEM) to see which roles own it

03 Each role shows salary range, certifications, career progression, and day-in-life

04 Career Transitions tab shows exactly how to move into or out of each role

A Day in the Life

What these roles actually look like, hour by hour.

SOC Analyst

8AM: Review overnight alerts in Splunk
10AM: Investigate phishing reported by employee
1PM: Update detection rules
3PM: Shift handoff briefing

GRC Analyst

8AM: Collect evidence for SOC 2 audit
10AM: Review vendor risk questionnaire
1PM: Update control mapping spreadsheet
3PM: Draft policy update for change management

Pen Tester

8AM: Scope and plan external assessment
10AM: Run automated scan + manual recon
1PM: Exploit discovered vulnerability
3PM: Document findings for client report

Cloud Security Engineer

8AM: Review IAM policy changes in AWS
10AM: Investigate S3 bucket misconfiguration
1PM: Build Terraform module with guardrails
3PM: Update CI/CD pipeline security checks

Incident Responder

8AM: (On-call) Ransomware alert triggered
10AM: Isolate affected endpoints, begin forensics
1PM: Coordinate with legal and leadership
3PM: Write incident report and lessons learned

AI Governance Analyst

8AM: Review new AI use case request form
10AM: Assess vendor AI model against NIST AI RMF
1PM: Update AI risk register, flag bias findings
3PM: Draft AI acceptable use policy section

Salary Reality by Role Family

What to expect at entry and mid-level across each path

Role Family	Salary Range (US)	Notes
Builders (Security Engineer)	$110K - $160K	High demand, mid-career growth strong
Breakers (Penetration Tester)	$100K - $145K	Competitive to enter, scarce openings
Defenders (SOC Analyst)	$75K - $120K	Most entry-level openings, clear ladder
Governors (GRC Analyst)	$80K - $115K	Huge volume of openings, underrated path
Cloud Security Engineer	$120K - $175K	Hardest to fill, highest ceiling

Digital Handout

If you don’t use or don’t want to use Claude account.

You won’t be able to save your results

If you can use Claude account.

Ability to save and email.

HANDS-ON

Exercise 1: Find Your Role Family

10 minutes | Use your handout. 8 scenarios. Be honest, not aspirational.

Each scenario has 4 responses. Circle the one that sounds most like you.

1 Your mom calls: "I clicked a link and now my screen is weird."

A) Remote in and rebuild her laptop properly B) Ask what she clicked so you can reverse-engineer it C) Walk her through disconnecting from WiFi immediately D) Set up a family password policy and make everyone sign it

2 You find a $20 bill on the ground at BSides. What now?

A) Build a lost-and-found tracking app B) Check the serial number for counterfeiting signs C) Turn it in to the front desk and monitor if someone claims it D) Write a "found property" incident report with time and location

3 Your team's Slack bot just started posting gibberish at 3 AM.

A) SSH in, check the code, patch the vulnerability B) Try to reproduce it and figure out who triggered it C) Kill the bot, check logs, alert the team D) Document the incident timeline and update the bot approval policy

4 Netflix asks "Are you still watching?" Your reaction:

A) "I could build a better recommendation engine" B) "I wonder if I can bypass this timeout" C) "Something is monitoring my behavior and I don't like it" D) "Is this compliant with their privacy policy?"

5 You find an unlocked laptop in a coffee shop. Owner is gone.

A) Check if full disk encryption is enabled (for science) B) Test how much sensitive data is exposed before anyone notices C) Guard it until the owner returns, note what was exposed D) Leave a sticky note: "Lock your laptop. Here's how to enable auto-lock."

6 Your company just bought a shiny new AI chatbot. You think:

A) "I need to see this architecture diagram" B) "I bet I can make it say something it shouldn't" C) "What happens when it hallucinates during an incident?" D) "Did anyone check if this meets our data retention policy?"

7 A recruiter DMs you: "We have a great cybersecurity opportunity!"

A) "What's your tech stack and cloud provider?" B) "Do you have a bug bounty program?" C) "What does your SOC team structure look like?" D) "What compliance frameworks do you maintain?"

8 You're assembling IKEA furniture. Your approach:

A) Ignore instructions, engineer it from the picture on the box B) Test which pieces can bear weight before assembling C) Follow each step carefully, alert someone if a piece is missing D) Read ALL instructions first, inventory every piece, THEN start

HANDS-ON

Your Results: Which Family Are You?

Count your A's, B's, C's, and D's. Your highest letter is your primary family.

MOSTLY A's — **BUILDERS**

You want to create, architect, and engineer. You see problems and think "I can build something better." You'd thrive as a Security Engineer, Cloud Security Engineer, DevSecOps, or Security Architect.

MOSTLY B's — **BREAKERS**

You want to probe, test, and find what's broken. You think "I wonder what happens if..." You'd thrive as a Pentester, Red Teamer, Bug Bounty Researcher, or AI Red Teamer.

MOSTLY C's — **DEFENDERS**

You want to protect, monitor, and respond. You think "Something's not right here." You'd thrive as a SOC Analyst, Incident Responder, Detection Engineer, or Threat Hunter.

MOSTLY D's — **GOVERNORS**

You want to organize, assess risk, and ensure accountability. You think "What's the process?" You'd thrive as a GRC Analyst, IT Auditor, Compliance Manager, or AI Governance Specialist.

Tied? You're a hybrid. That's an advantage, not a problem. Most security leaders draw from multiple families.

Career Migration Paths

Where you start today, where each path leads, and which certs get you there.

WHERE YOU ARE		WHERE YOU'RE GOING	CERTS THAT GET YOU THERE
Help Desk / IT Support	→	SOC Analyst, Security Ops, IAM Engineer	CompTIA Security+, ISC2 CC (free), CySA+
Sysadmin / IT Ops	→	Security Engineer, Cloud Security, Patch/Config Mgmt	AWS Security Specialty, AZ-500, CompTIA Security+
NOC Analyst	→	Detection Engineer, Incident Responder, Threat Hunter	CySA+, BTL1, GIAC GCIH
Developer / DevOps	→	AppSec Engineer, DevSecOps, Product Security	CSSLP, GWEB, AWS DevOps Professional
Compliance / Legal	→	GRC Analyst, IT Auditor, Compliance Manager	CISA, CRISC, CISM, ISO 27001 Lead Auditor
Data / AI/ML	→	AI Security Engineer, AI Governance Specialist	CompTIA AI+, ISO 42001, NIST AI RMF

Start with ONE foundational cert. Align to your target role family. Build experience alongside certifications, not instead of them.

How to Actually Get In

When employers want experience you do not have yet

Adjacent Roles First

IT roles at defense contractors, financial services, or healthcare. These on-ramps have real security exposure.

Build in Public

Labs, GitHub, write-ups, CTFs. Employers are shifting to competency-based hiring: portfolios over degrees.

Community & Networking

BSides, local ISSA/ISACA chapters, Discord servers. Most jobs are filled through relationships, not applications. Volunteer.

Skills Are Your Edge

Explaining risk to non-technical people is rare and valuable. Communication wins interviews. Sell your transferable skills.

Strategic Credentialing

Start with one foundational cert (Sec+, CySA+), Align certs to your target role family, Build experience alongside certifications

Target Emerging Roles

AI governance, cloud security, Zero Trust, DevSecOps, IAM. Less competition, growing demand.

The SOC Analyst Is Not Dying.
It Is Evolving.

Shrinking

Manual alert triage
Copy-paste ticket workflows
First-pass log review
Basic indicator lookups

Growing (+40%)

Detection engineering & custom rules
AI-assisted threat hunting
Automation development (SOAR playbooks)
AI decision auditing & validation

The Evolution Path

SOC Analyst > Detection Engineer > Threat Hunter > AI SOC Supervisor > SOC Manager

Skills to Add Right Now

AI decision auditing | SOAR playbook development | Detection rule writing (Sigma, YARA) | Prompt engineering for security | Business context analysis

Only 2% believe AI will fully replace cybersecurity professionals. But 52% agree entry-level repetitive work is changing.

Your AI Cybersecurity Learning Path

AI is the #1 most-needed skill in cybersecurity (ISC2 2025). Here's how to start.

FUNDAMENTALS

FREE

- Anthropic Prompt Engineering (9 chapters)
- DeepLearning.AI LLM courses
- fast.ai Practical Deep Learning
- Google Cloud Security AI on Coursera

AI SECURITY

FREE

- OWASP Top 10 for LLM Applications
- MITRE ATLAS (AI threat framework)
- Microsoft Security Copilot training
- AWS AI/ML Security learning path

AI GOVERNANCE

MID $$

- NIST AI RMF Playbook
- ISO 42001 (AI Management)
- EU AI Act compliance
- ISC2 AI for Cybersecurity course
- IBM Generative AI for Cyber Professionals

HANDS-ON + CERTS

$$$

- CompTIA SecAI+ (launched Feb 2026)
- SANS SEC411: AI Security
- Certified AI Security Professional (CAISP)
- HackTheBox AI labs
- Claude Code / AWS Bedrock

You don't need a PhD. You need to understand how AI tools work, query them effectively, validate outputs, and govern their use.

GRC: The Most Underrated Career in Cybersecurity

Why GRC?

- No coding required (process, policy, frameworks)
- Diverse backgrounds are an advantage (accounting, legal, HR, PM)
- Most CISOs come from GRC/risk backgrounds
- 34,000+ GRC job postings in the U.S. (2023)
- Interest spiked ~1,000% on LinkedIn over 5 years
- AI Governance Specialist: $119K-$215K (new in 2025)
- LEAST competition in public content. A single policy template on GitHub puts you ahead of 95% of candidates.

The Career Ladder

GRC Analyst	$65K-$85K
Compliance Manager	$95K-$130K
Director of GRC	$140K-$180K
VP / Head of Security	$170K-$250K
CISO	$220K-$483K+

This is my career path. I live this every day.

Frameworks to start with: SOC 2 (SaaS), ISO 27001 (global), NIST CSF (U.S. federal). Learn one deeply, then expand.

How to Build in Public (By Role Family)

The single best thing you can do to stand out: show your work where hiring managers can find it.

BUILDERS

Terraform modules with security guardrails

CI/CD pipeline security templates

IaC scanning configs

Open-source security tool contributions

Share on: GitHub, Dev.to, LinkedIn

BREAKERS

CTF writeups (TryHackMe, HackTheBox)

Bug bounty reports (disclosed)

Custom tool development

Vulnerability research blogs

Share on: GitHub, Medium, personal blog, LinkedIn

DEFENDERS

Sigma / YARA detection rules

SOAR playbook templates

SIEM dashboard configs

Investigation writeups (sanitized)

Share on: GitHub, LinkedIn articles

GOVERNORS

Policy templates (AUP, IR, access control)

Control mapping sheets (PCI to NIST CSF)

Risk register templates

Compliance narrative examples

Audit prep checklists

Share on: GitHub, LinkedIn

GRC people: you have the LEAST competition in public content. A single policy template on GitHub puts you ahead of 95% of candidates.

Real Examples: Study, Fork, Contribute and/or Build your own

These are actual repos and projects people built. You can do the same.

BUILDERS

nozaq/
terraform-aws-secure-baseline

CIS + AWS best practices as Terraform. 2K+ stars.

naman16/
AWS-Security-Guardrails-TF

Auto-generates secure Terraform from Checkov/Prowler rules.

cloudposse/
terraform-aws-security-hub

Production-ready IaC for AWS Security Hub.

BREAKERS

Ignitetechnologies/
TryHackMe-CTF-Writeups

Massive collection of THM walkthrough writeups.

Ignitetechnologies/
HackTheBox-CTF-Writeups

HTB machine writeups sorted by difficulty.

Kelvinkimotho/
CTF-Writeups

Personal archive: exactly what YOUR repo could look like.

DEFENDERS

SigmaHQ/
sigma

3,000+ detection rules. THE repo for Sigma. Contribute here.

luduslibrum/
awesome-playbooks

1,347 IR playbooks across 12+ SOAR platforms.

socfortress/
Playbooks

SOC IR playbooks per NIST 800-61. Fork and customize.

GOVERNORS

JupiterOne/
security-policy-templates

Full policy set: HIPAA, NIST, SOC 2, PCI, FedRAMP.

securitytemplates/
sectemplates

Bootstrap templates for security programs + bug bounty.

counteractive/
incident-response-plan-template

Complete IR plan in Markdown. Fork it tonight.

All free. All on GitHub. Pick one tonight, study the structure, then build your own version.

BREAK

5 minutes

Stand up. Stretch. Find someone you haven't spoken to yet.
Introduce yourself.

The Skills Nobody Tests For (But Everyone Needs)

ISC2 2025: 51% of hiring managers say nontechnical skills are becoming MORE important with AI.

Communication

Explain a CVE to a VP in 30 seconds. Write a finding so a developer actually fixes it.

Critical Thinking

Don't blindly accept AI output. Question assumptions. Validate evidence. This is your superpower.

Business Context

Know what the company does, how it makes money, and which systems matter most. Security exists to protect the mission.

Collaboration

Security is a team sport. You'll work with developers, legal, HR, executives. Learn to influence without authority.

Adaptability

The tooling you learn today will be different in 2 years. The ability to learn fast is the only permanent skill.

Resume Refactoring: Before & After

Every bullet point should answer: "So what? What was the impact?"

DEFENDERS

Before: Monitored security alerts

After: Monitored 500+ endpoints using CrowdStrike, escalating 15 incidents/week with 98% accuracy SLA

BUILDERS

Before: Worked on cloud security

After: Built Terraform modules with IAM guardrails deployed across 12 AWS accounts, reducing misconfigurations 40%

BREAKERS

Before: Did penetration testing

After: Conducted 15 external assessments in Q3, identifying 23 critical vulnerabilities and reducing remediation time 35%

GOVERNORS

Before: Helped with compliance

After: Led SOC 2 Type II evidence collection across 45 controls, achieving zero exceptions in annual audit

The pattern: Action Verb + Scope/Scale + Tool/Method + Quantified Result

HANDS-ON

Exercise 2: Live Resume Teardown

25 minutes | This is where the real work happens.

Phase 1: Group Teardown (10 min)

- Volunteer resume on screen
- We'll score it on our rubric together:
  - Does the summary name a target role?
  - Are tools listed prominently?
  - Do bullets use impact framing?
  - Is it ATS-friendly (single column, no graphics)?

Phase 2: Peer Review (15 min)

- Pair up with someone nearby
- Pull up your resume on your phone/laptop
- 4 min reading, 4 min discussing each
- Answer for each other:

1. What role does this person seem to be targeting?
2. What's the strongest bullet?
3. What could be improved?
4. Is a skill or experience buried?

No resume yet? That's OK. Pair with someone who has one and practice giving feedback. That's a skill too.

If your resume says "Responsible for..." change it right now. Action verbs only.

Cracking the Interview

I hire people globally. This is exactly what I look for.

Before the Interview

- Tailor your resume for EVERY application (Application Tracking system (ATS) keywords matter)
- Research the company's security posture (breaches, compliance, tech stack)
- Prepare 3 stories using STAR format (Situation, Task, Action, Result)
- Have questions ready that show you researched the role
- Single column resume, standard fonts, no graphics (ATS-friendly)

During the Interview

- Lead with enthusiasm. Energy > perfection.
- "I don't know, but here's how I'd find out" > making something up
- Show your thought process, not just answers
- AI rule: Never use AI during the interview
- Send a thank-you email within 24 hours (specific, not generic)
- Follow up once after 1 week. Then let go.

Run your own race. Control what you can. Go all in, then let go.

HANDS-ON

Exercise 3: LinkedIn Quick-Fire Audit

10 minutes | Pull out your phone right now.

Score yourself. Be honest.

1. Professional headshot (not a crop from a party photo)
2. Custom headline (not just "Student at ___")
3. Banner image (free Canva templates work fine)
4. About section filled in (3+ sentences about what you want to do)
5. 3+ cybersecurity-relevant skills listed
6. Education section completed
7. At least 1 experience or project listed
8. Customized URL (linkedin.com/in/yourname)
9. 50+ connections
10. Engaged with cybersecurity content in the last 30 days

8-10: Solid. Help your partner. **5-7:** Decent. Fix 2 things today. **Below 5:** Your homework starts now.

Now swap phones with your partner. Each give one specific improvement suggestion.

The Mindset

Inspired by The Pursuit of Happyness: "Don't ever let somebody tell you that you can't do something."

Run Your Own Race

Stop comparing timelines. Someone got hired faster? Different race. Your only competition is who you were last month.

Control the Controllables

You control preparation, energy, follow-up. You don't control who else applied, hiring timelines, or internal politics.

Go All In, Then Let Go

Prepare like it's the only opportunity. Then release the outcome. Let go of what you can't control.

Some companies: ~1M applications/year for ~350 hires. Written interviews, psychometric tests, 6-10 rounds, 2-6 months.

They will test your patience, your ego, and your endurance. Be ready.

Imposter Syndrome

"I'm not good enough. Everyone else knows more than me. I don't belong here."

Why Cybersecurity Makes It Worse

- The field is impossibly broad (nobody knows it all)
- Job postings ask for everything
- Technology changes faster than you can learn
- Social media highlights experts, not learners
- Certifications create a "never enough" treadmill

How to Move Through It

- Name it. "I'm feeling imposter syndrome" neutralizes it.
- Track your wins (a simple doc of things you've learned/done)
- Compare to your past self, not to Twitter/LinkedIn highlights
- Find your 3: mentor, peer, mentee
- Remember: the person interviewing you also Googles/Chatgpt/Claude things

I have a dual Master's, four certifications, 16 years of experience, and I still feel imposter syndrome. You're not alone.

HANDS-ON

Exercise 4: Your 30-Day Career Sprint

10 minutes | You're leaving with a plan and an accountability partner.

Fill out your plan. Use the worksheet. Be specific.

1. My Cybersecurity Target

Target role:
Why this role:
Biggest gap I need to close:

2. Three 30-Day SMART Goals

LEARN: Complete ___ by [date]
BUILD: Create ___ by [date]
NETWORK: Reach out to ___ people by [date]

3. My Accountability Partner

Partner name:
Contact info:
Check-in date (2 weeks from today):

Turn to someone you haven't spoken to. Introduce yourself, exchange contact info, and schedule a 15-minute check-in for April 24.

Key Takeaways

- The field is growing, but entry-level is competitive. Be strategic, not desperate.
- Find your role family (Builders/Breakers/Defenders/Governors) and go deep.
- AI isn't replacing you. But someone who knows AI will outcompete someone who doesn't.
- GRC is the most underrated, least crowded, highest-ceiling path in cyber.
- Build in public. Show your work. The best resume is one hiring managers find before you apply.
- Certifications open doors. Skills keep you in the room. Relationships change your career.
- Your path will not look like mine. That is the point.

About the Facilitator

Rupinder Pal Singh

CISSP | CISM | CISA | CRISC

- 16+ years across cybersecurity, audit, risk, compliance, and privacy
- Manager, Information Security at NiCE (cloud contact center)
- Leads global teams across the US, UK, EU, Australia, and Philippines
- Programs: FedRAMP, SOC 2, PCI DSS, ISO 27001, ISO 42001
- Dual Master's: Cybersecurity & Clinical Mental Health Counseling
- ISACA contributor: exam item writer, manual reviewer, scholarship judge

LinkedIn — Rupinder Pal Singh

linkedin.com/in/rupinderpalsinghsecurity/

Connect for ongoing career conversations, resources, and job leads in cybersecurity.

Rupinder Pal Singh

Manager, Information Security | NiCE

Slides

Questions, conversations, and connections are all welcome.

Thank you.

OPEN INVITATION

Come find me after. Whether you are just starting out, pivoting careers, or trying to level up. I am here for the conversation.

Interactive Defense in Depth Map

All 40+ roles, salary ranges, certifications, career paths, and day-in-life details. Explore at your own pace.